A Contrastive-Pairs Benchmark for Structural Manipulation

Protocol and Pilot — Does a Model Detect Rhetorical Structure, or Only Vocabulary?

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Politomorphism Engine — Defensive Political Anthropology Layer

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1. Why this benchmark exists

The working papers in this programme established that a contextual model classifies PropXplain's propaganda labels better than a lexical one. Independent audits raised the deeper question those papers could not answer: does any of these models detect manipulation as a structure, or do they merely key on vocabulary? A model that flags “propaganda” because the text contains charged words has learned a lexical correlate, not the rhetorical mechanism. This benchmark is built to separate the two.

The method is contrastive pairs. Each item is a pair of short texts on the same topic, using as nearly as possible the same words, where one member carries a manipulative rhetorical structure — a presupposition, a manufactured equivalence, a false dilemma, a whataboutist deflection, a loaded rhetorical question, an implied causation — and the other states the same content plainly, without that structure. Because the vocabulary is matched, a model that scores the manipulative member higher must be responding to something beyond the words. The primary metric is pair accuracy: the fraction of pairs in which the manipulative member receives the higher propaganda score. A purely lexical model, on perfectly matched pairs, should sit near 0.5; a model sensitive to structure should sit well above it.

2. What the benchmark measures, and what it does not

This benchmark measures sensitivity to rhetorical structure, holding vocabulary approximately fixed. It does not measure whether a text “is propaganda” in any political or legal sense, whether it would persuade a real audience, or whether the construct “manipulation” is validly operationalized at all. It is one instrument in a construct-validity programme, not a verdict. A high pair accuracy is evidence that a model represents structure; it is not evidence that the model, or the corpus labels it was trained on, captures manipulation as it operates in the world. The ethical boundary of the wider programme applies here too: an instrument that scores rhetorical structure must never be used to adjudicate which political positions are legitimate.

3. Construction rules for a pair

A valid pair satisfies five rules. First, same topic: both members concern the same event, actor, or policy. Second, matched vocabulary: the two members should share as many content words as possible; the structural difference, not the word list, should distinguish them. Third, one structural device: the manipulative member introduces exactly one named technique, so that a failure can be attributed. Fourth, a plain control: the control states the same content as an open, attributed, or multi-option claim, removing the device without adding a new one. Fifth, a written rationale: each pair records, in one sentence, why the manipulative member is manipulative and why the control is not.

Pairs are labelled by technique so that pair accuracy can be reported per technique, which is where the interesting structure lives.

4. Metrics and controls

The headline metric is pair accuracy with a binomial confidence interval. Two controls guard against misreading it. The first is a lexical-overlap control: for each pair we report the Jaccard overlap of the two members' word sets. If overlap is low, the pair is poorly matched and a lexical model can win on vocabulary alone, so pair accuracy on low-overlap pairs is not clean evidence of structure sensitivity; the benchmark should report accuracy conditional on overlap, or restrict to high-overlap pairs. The second is the baseline-versus-contextual comparison: the key quantity is not either model's absolute pair accuracy but the margin by which a contextual model exceeds the lexical baseline on well-matched pairs. That margin, if positive and robust, is the evidence that context buys structural sensitivity — the claim the corpus studies could not establish.

5. Pilot results (Version 0.2, vocabulary-matched)

A first 24-pair pilot (v0.1) proved the method but was too loosely matched on vocabulary (mean member Jaccard 0.21), so a lexical model could still win on words. Version 0.2 rewrote the set as 30 pairs — five per technique across presupposition, manufactured equivalence, false dilemma, whataboutism, loaded rhetorical question, and implied causation — holding the two members as close as possible in wording and changing only the structure, which raised mean member Jaccard to 0.53. Both the TF-IDF + logistic-regression baseline and the fine-tuned DistilBERT (trained on the clean PropXplain split) scored every member; Table 1 reports pair accuracy by technique for each on v0.2.

Technique	Pairs	Baseline	Transformer
presupposition	5	1.00	1.00
manufactured equivalence	5	0.80	1.00
false dilemma	5	0.40	1.00
whataboutism	5	1.00	1.00
loaded rhetorical question	5	0.00	0.80
implied causation	5	0.80	1.00
ALL	30	0.667 [~.50,.84]	0.967 [~.90,1.0]

Table 1. Lexical baseline vs. fine-tuned transformer on the 30-pair v0.2 set, by technique (fraction of pairs where the manipulative member scored higher). Mean member Jaccard overlap 0.53, so vocabulary is substantially matched.

On vocabulary-matched pairs the result is clean and points one way. The transformer reaches 0.967 overall against the baseline's 0.667, and the two confidence intervals do not overlap (baseline approximately [0.50, 0.84]; transformer [0.90, 1.00]), so on this set the margin is statistically distinguishable, not merely suggestive. More telling than the overall number is where the gap lives: on the two techniques where the lexical baseline failed structurally once vocabulary was matched — loaded rhetorical questions (baseline 0.00, transformer 0.80) and false dilemma (0.40 to 1.00) — the transformer recovers almost everything. Because the two members of each pair share most of their words, that recovery cannot be attributed to vocabulary; it is evidence that the contextual model is sensitive to rhetorical structure, which is the central claim the corpus-level studies could not establish.

Three honest limits keep this a strong pilot rather than a settled finding. The set is still small (30 pairs), and both the pairs and their manipulative/control labels were written by the author alone — so the

labels have not been checked by an independent annotator, and the pairs and the model come from the same programme. Mean overlap of 0.53 is much improved but not identity, so a residue of vocabulary difference remains, concentrated on techniques like loaded rhetorical questions whose control phrasing ("people are asking whether …") differs in form. And this is one model at one seed. The result justifies the next steps, not a general claim that transformers "detect manipulation."

6. The experiment this sets up

Once the pairs are tightly matched, the decisive comparison is between the lexical baseline and the contextual transformer on the same pairs. If the transformer's pair accuracy exceeds the baseline's on high-overlap pairs — that is, when vocabulary cannot explain the difference — that margin is the first direct evidence that context captures rhetorical structure, not just words, which is the claim the corpus-level studies in this programme could not establish. The script 06_transformer_pairs.py produces the transformer's per-member scores; evaluate_pairs.py computes pair accuracy for either model. The benchmark then extends naturally: more techniques, more pairs per technique, independent human annotation of the pairs (so the labels are not the author's alone), and stronger models.

7. Next steps

Version 0.2 has delivered the first clean, vocabulary-matched signal. In order, the remaining steps are: obtain at least one independent annotator's judgments on the 30 pairs and report inter-annotator agreement, so the labels are not the author's alone; expand to roughly ten pairs per technique and re-check that the margin holds; tighten the control phrasing on loaded rhetorical questions so even that residue of vocabulary difference is removed; repeat the transformer run across several seeds for a confidence interval on the margin; and only then write this up as a short third paper. Construct validity is earned pair by pair and annotator by annotator, not claimed. All materials — both pair sets, the scoring harness, per-member scores for baseline and transformer, and this protocol — are in the repository so that others can inspect, criticize, and extend them.

Materials

Pilot pairs, the evaluation harness (evaluate_pairs.py), the transformer scorer (06_transformer_pairs.py), and this protocol are maintained at github.com/politomorphism/politomorphism-defense.